Chapter 7
Structure and Preparation of Alkenes: Elimination Reactions

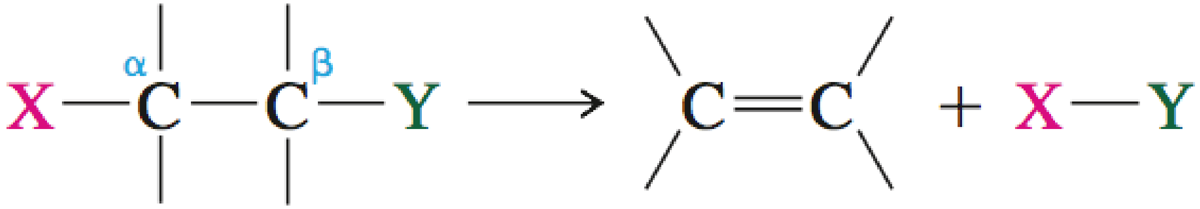
Additional Recommended Problems: Chp7: 11,12,25,26,31,38,39,41,42,43,45,49,50-55

CH 7 Learning Objectives

- Correctly name alkenes, using the E/Z and cis-trans systems.
- Predict the products of E1 and E2 reactions, including stereochemistry.
- Use Zaitsev's Rule to predict which structural isomer will predominate in E2 or E1 reactions.
- When a halocarbon reacts, identify when S_N2 or E2 reactions occur, or when S_N1 or E1 reactions will occur, and predict the major products.
- Draw mechanisms for any of S_N1, S_N2, E1, or E2 reaction
- Rank the relative rates of substitutions or eliminations reactions, based on differences in substrate, base/nucleophile, leaving group, or solvent.
- Predict whether a reaction will be first-order or second-order
- When possible, predict predominance of substitution or elimination
- Identify reactants that could product target chemical products
- Design multi-reaction synthesis design sequences to convert hydrocarbons to more highly functional derivatives

Synthesis of Alkenes

- A **β-elimination** reaction involves the loss of X and Y from adjacent carbon atoms with formation of C=C



- One group is *electronegative* (X) and the other is *electron-releasing* (Y)
- For example, X is a good leaving group and Y is an acidic hydrogen atom
- Possible for atoms other than carbon as well!

Dehydrogenation

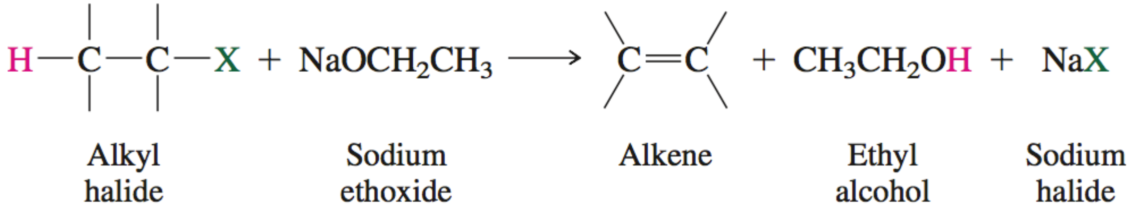
- Industrially, ethylene and propylene are prepared by the metal-assisted elimination of H_2

$$\underset{\text{Ethane}}{CH_3CH_3} \xrightarrow{750°C} \underset{\text{Ethylene}}{H_2C{=}CH_2} + \underset{\text{Hydrogen}}{H_2}$$

$$\underset{\text{Propane}}{CH_3CH_2CH_3} \xrightarrow{750°C} \underset{\text{Propene}}{CH_3CH{=}CH_2} + \underset{\text{Hydrogen}}{H_2}$$

Most Common Eliminations

- Dehydrogenation is impractical for synthesis of most alkenes; instead, a substrate containing a good leaving group (or precursor to one) is used
- **Dehydrohalogenation** involves elimination from H–C–C–X

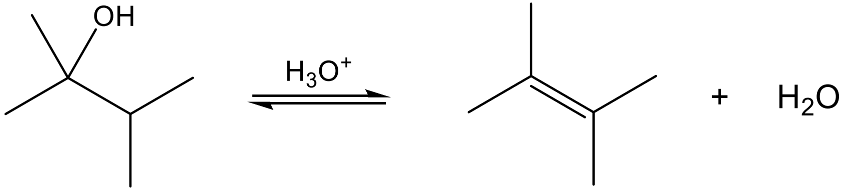


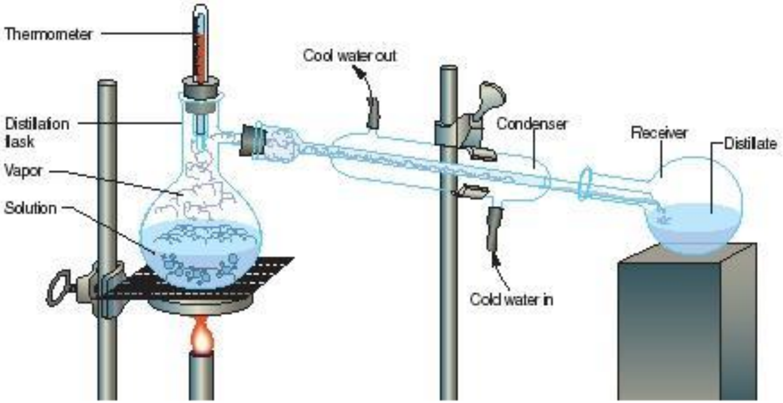
- **Dehydration** involves elimination from H–C–C–OH

$$\underset{\text{Ethyl alcohol}}{CH_3CH_2OH} \xrightarrow[160°C]{H_2SO_4} \underset{\text{Ethylene}}{H_2C{=}CH_2} + \underset{\text{Water}}{H_2O}$$

E1 - Dehydration

- Mechanism resembles E1 (no S_N1)
- Acid catalyst (conc. H_2SO_4, H_3PO_4)

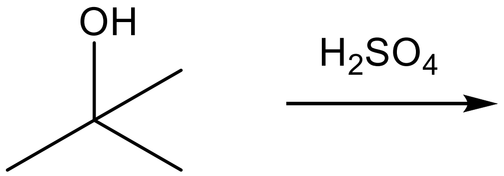




E1 Mechanism Dehydrogenation

tert-Butyl alcohol

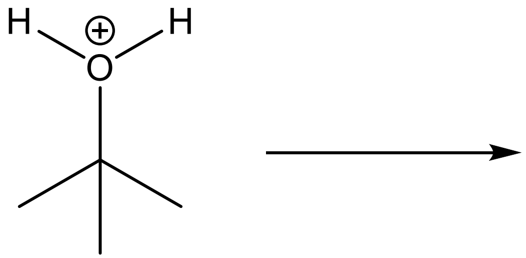
- Step 1: Protonation of the alcohol



E1 Mechanism Dehydrogenation -1

tert-Butyl alcohol

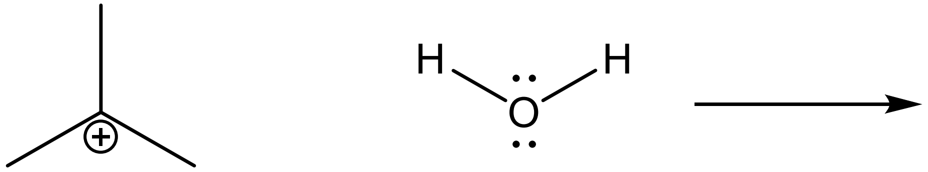
- Step 2: Dissociation of *tert*-butyloxonium ion to form carbocation



E1 Mechanism Dehydrogenation -3

tert-Butyl alcohol

- Step 3: Deprotonation of the β-hydrogen by water to form the alkene



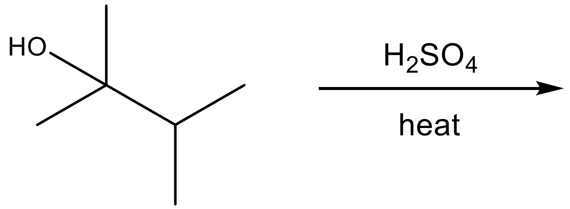
What if More than One β-hydrogen?

- **Zaitsev's Rule:** In alcohol dehydrations, the most substituted alkene is formed preferentially

- Why? Alkenes are *stabilized* by electron-releasing alkyl substituents

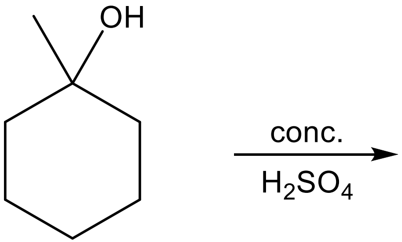
- Ultimately, Zaitsev's rule reflects the idea that the major product in alcohol dehydrations is the most thermodynamically stable product

Zaitsev’s Rule Example (Regioselectivity)

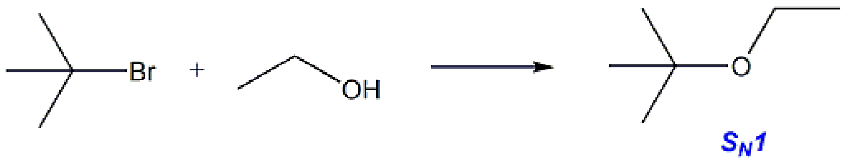


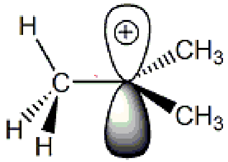
Dehydration Example

- Draw a mechanism for the following reaction and who both alkenes that are formed. Circle the major product.



Eliminations – E1





- S_N1 occurs with E1
- Carbocations can:

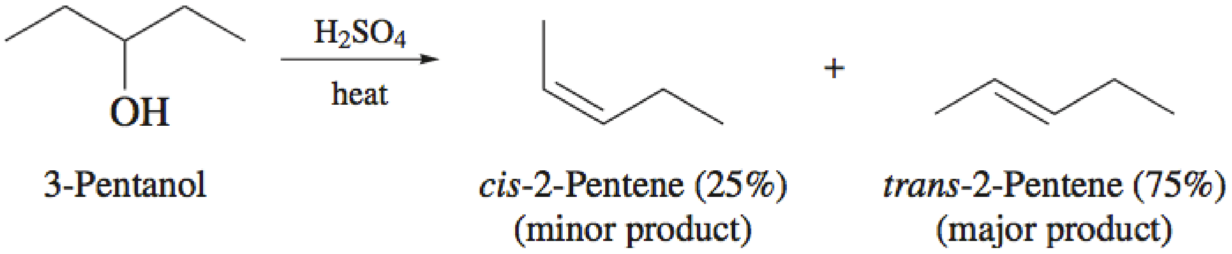
React with Nuc (S_N1)

Lose H^+ to form alkene (E1)

Rearrange and react further (S_N1/E1)

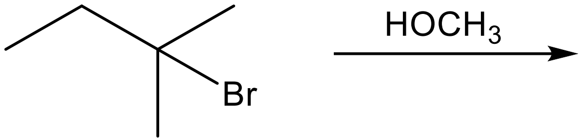
E1 Stereoselectivity

- Alcohol dehydrations can often yield either (*E*)- or (*Z*)-alkenes
- Based on the thermodynamic statement of Zaitsev's rule, we might expect the (*E*)-alkene to be formed preferentially
- Experimental results support this hypothesis



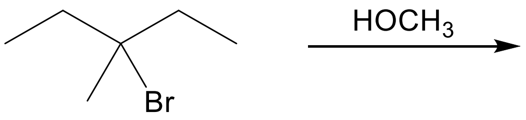
In general, the more stable *trans* or (*E*) isomer is formed preferentially in alcohol dehydrations.

Dehydrohalogenation E1 Example



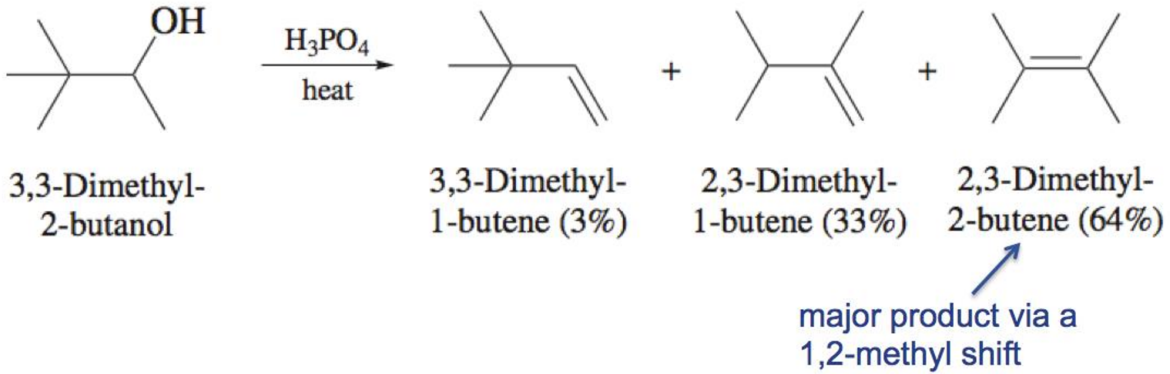
E1 Stereochemistry and Regiochemistry

- Draw all possible products for the following E1 reaction
- Of the products drawn, which is the most stable?

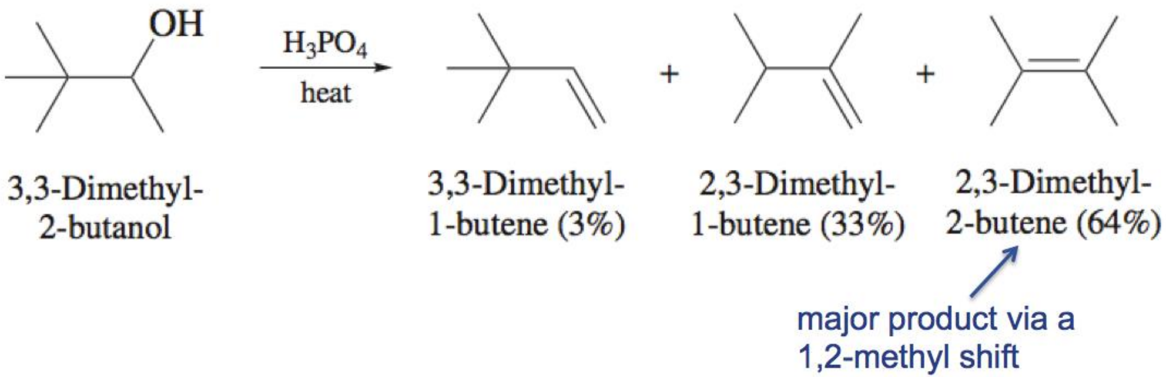


Rearrangements in E1 Reactions

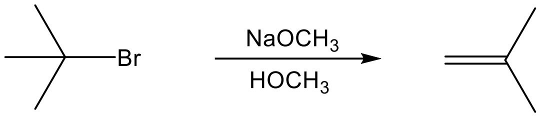
- Because E1 reactions involve a carbocation intermediate, rearrangements should be expected
- Rearrangement occurs when a more stable carbocation can be formed via migration of –H or $-\mathrm{CH_3}$ by one carbon
- Common for adjacent *tert*-butyl groups:

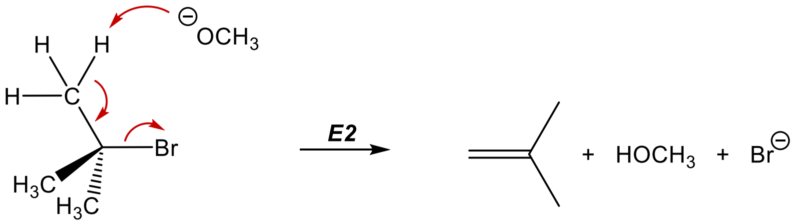


Rearrangements in E1 Reactions-1



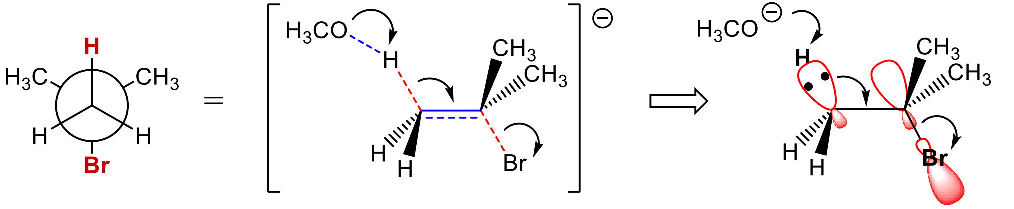
E2 Mechanism

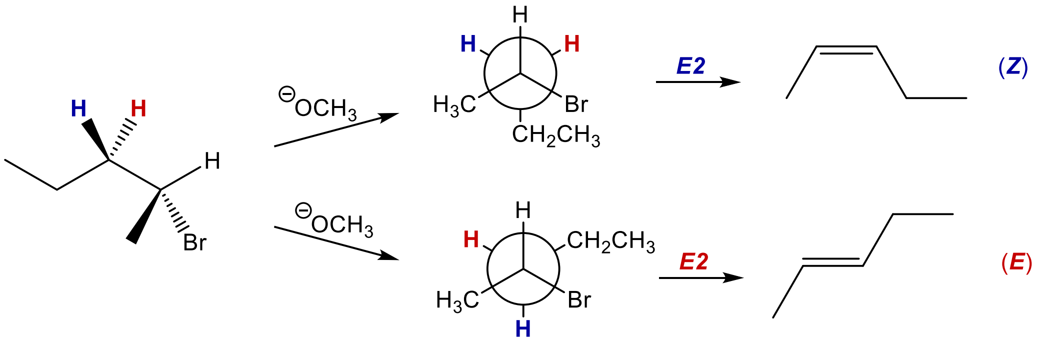




- TS requires coplanar arrangement (anti preferred)
- Reactivity: 3º > 2º > 1º
- Regioselective and stereoselective

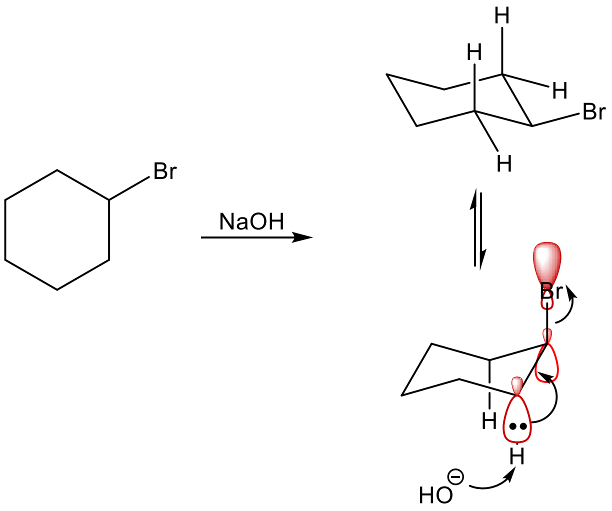
E2 Transition State





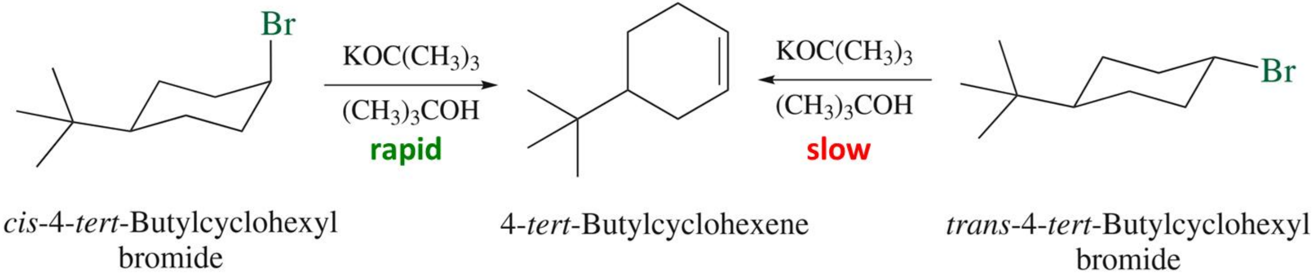
- H and X orbitals align to begin π-bond formation
- Anti-coplanar preferred (shown)
- Prearrangement leads to stereoselectivity

E2 in Cyclohexanes



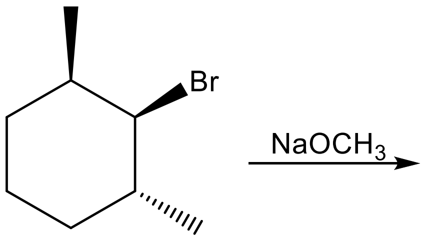
- H and X must adopt a *trans*-diaxial arrangement

Kinetics for Eliminations in Cyclohexanes Conformations

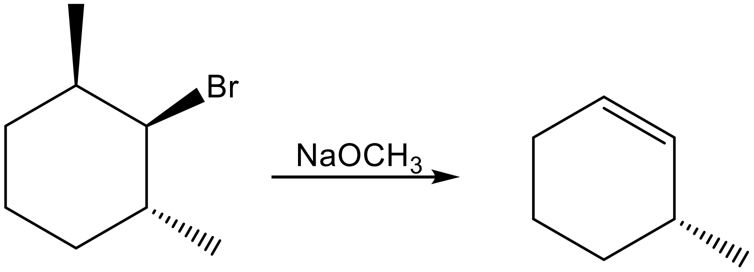


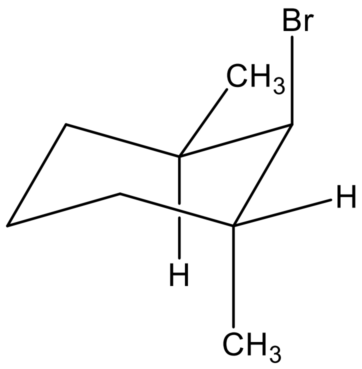
- An axial Br eliminates readily in the presence of strong base, but an equatorial Br does not

E2 in Cyclohexanes



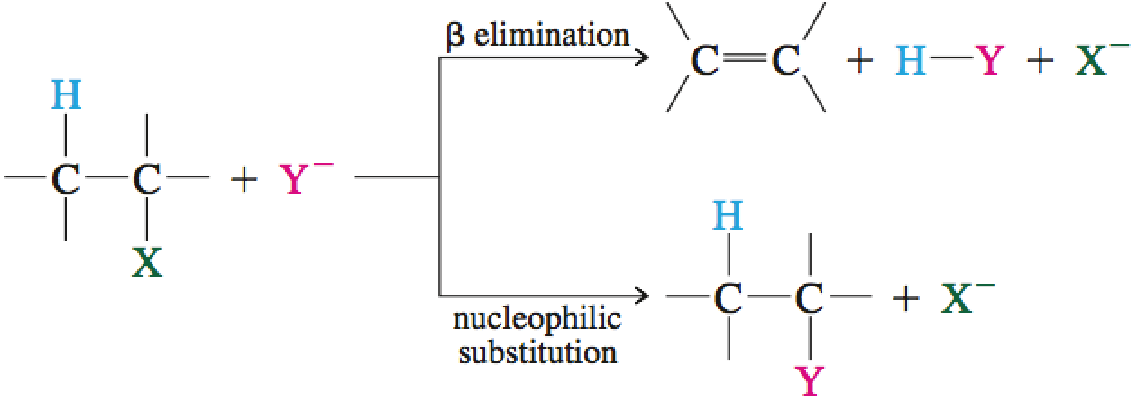
E2 in Cyclohexanes





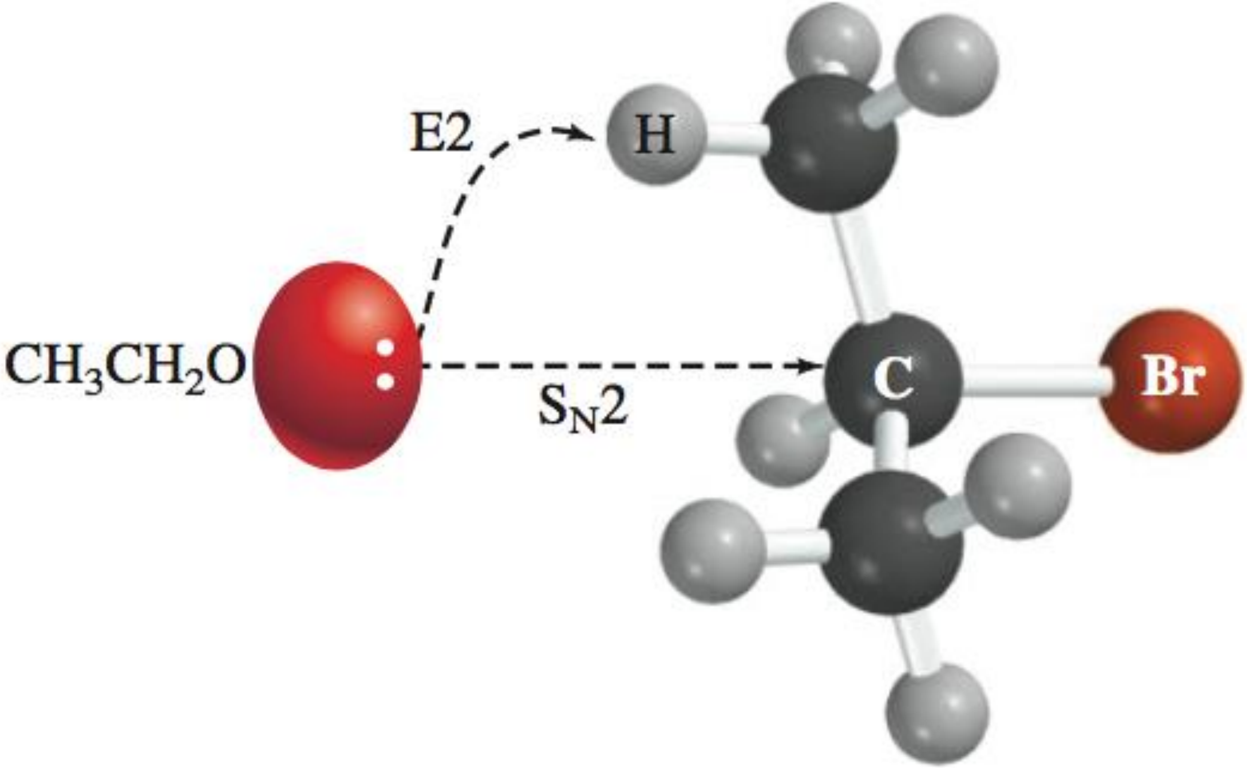
Competing Reactions

- Substitution and elimination involve the same substrates and reaction conditions

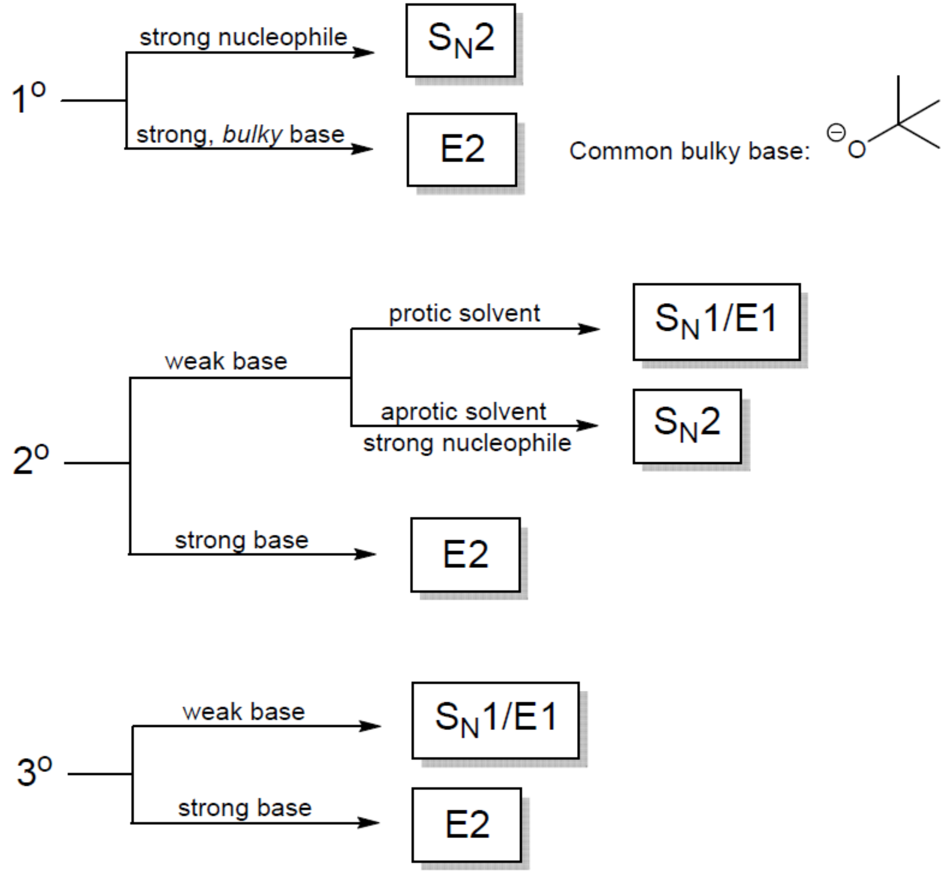


- Using previously discussed reactivity trends, we can predict whether substitution or elimination will occur given alkyl halide and Lewis base structures

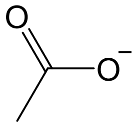
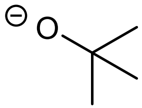
Brønsted or Lewis Base?



Flow Chart


1°
strong nucleophile
S_N2
strong, *bulky* base
E2
Common bulky base:
2°
weak base
protic solvent
$S_N1/E1$
aprotic solvent
strong nucleophile
S_N2
strong base
E2
3°
weak base
$S_N1/E1$
strong base
E2

Base/Nucleophile Chart

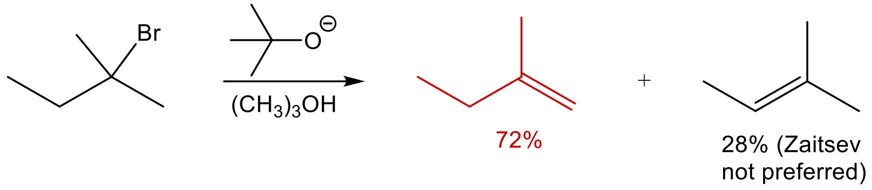
Strong nucleophile/weak base	Strong nucleophile/strong base	Strong bulky base	Weak nucleophile/weak base
Promote S_N2	Promote $S_N2/E2$	Promote E2	Promote $S_N1/E1$
N_3^- RS^- HS^- NO^- NC^- I^- Br^- 	^-OH $^-OCH3$ $^-OCH_2CH_3$		CH_3OH (MeOH) H_2O CH_3CH_2OH (EtOH)
Conjugate acids have pKa < 11	Conjugate acids have pKa about 16-18		Heat favors E1 Cold favors SN1

S_N2 vs. E2 pKa Table – Fill in on your own

Functional Group	Structure	Conjugate Base	pKa
Ammonia			
Alcohol			
Water			
Thiol			
Ammonium			
Hydrogen cyanide			
Carboxylic acid			
Hydrazoic acid			
HX			

Reaction Conditions

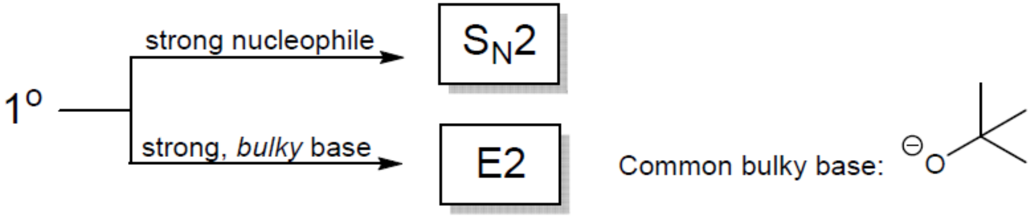
- Bulky base – promotes *E2* often with 1º R-X



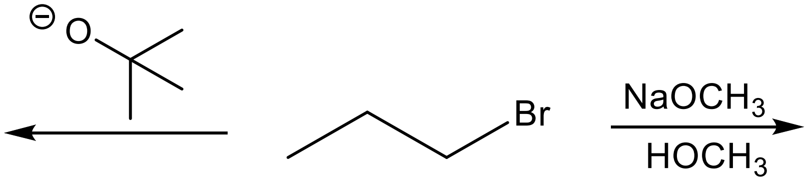
- Weak bases (Br^{1-}, I^{1-}, NC^{1-}, RS^{1-}, N_3^{1-}) – often promote S_N2 with 2º R-X

- Heating reaction increases amount of elimination product in S_N1/E1 mixtures ($-T\Delta S$ increases)

Flow Chart



1°
strong nucleophile
strong, *bulky* base
S_N2
E2
Common bulky base:
O



O
Br
$NaOCH_3$
$HOCH_3$

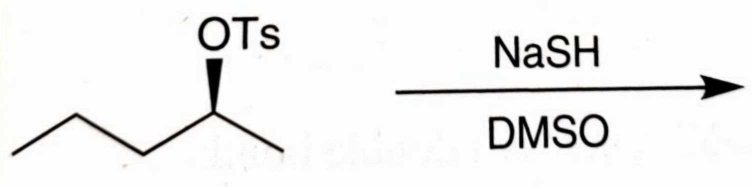
Reaction Conditions

- Bulky base – promotes *E2* often with 1º R-X

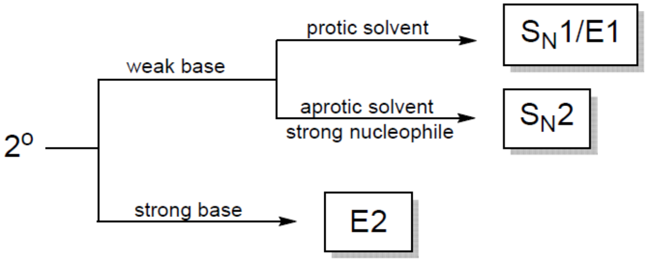
- Weak bases (Br^{1-}, I^{1-}, NC^{1-}, RS^{1-}, N_3^{1-}) – often promote S_N2 with 2º R-X

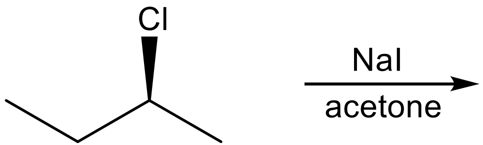
- Heating reaction increases amount of elimination product in $S_N1/E1$ mixtures ($-T\Delta S$ increases)

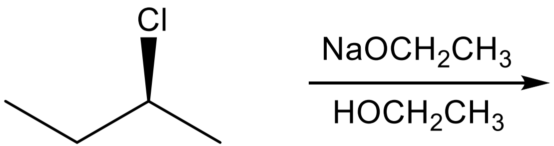
Draw the major product for the following reaction and identify the mechanism



Flow Chart

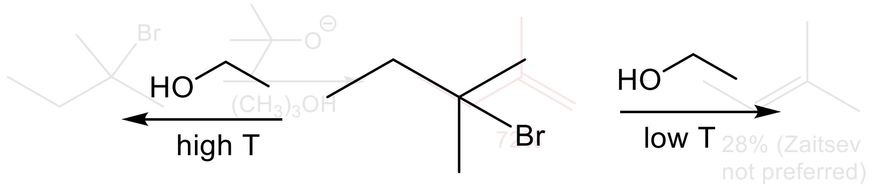

protic solvent
$S_N1/E1$
weak base
aprotic solvent
strong nucleophile
S_N2
2°
strong base
E2


Cl
NaI
acetone


Cl
$NaOCH_2CH_3$
$HOCH_2CH_3$

Reaction Conditions

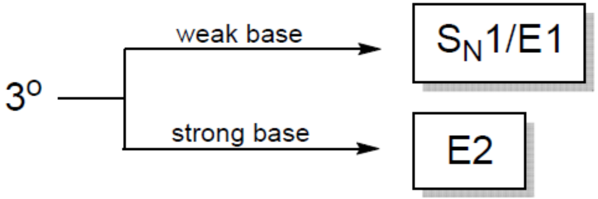
- Bulky base – promotes *E2* often with 1° R-X



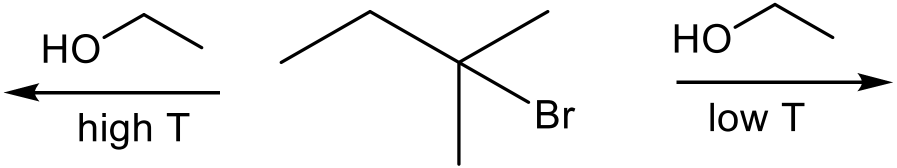
- Weak bases (Br^{1-}, I^{1-}, NC^{1-}, RS^{1-}, N_3^{1-}) – often promote S_N2 with 2° R-X

- Heating reaction increases amount of elimination product in $S_N1/E1$ mixtures ($-T\Delta S$ increases)

Flow Chart



Heat favors E1
Cold favors S_N1



Solvents in Organic Chemistry

- Solvents should be "unreactive" (affect rates)
- Protic (H-bonding) or Aprotic
- Polar (ε > 20) or Nonpolar (ε < 20)

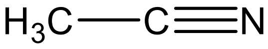
- S_N1 favors polar protic solvents (H_2O, ROH)
  - Can solvate ions and decrease nucleophilicity
  - Can stabilize carbocation in TS
- S_N2 favors polar aprotic solvents (CH_3CN, DMF, DMSO, acetone)

Solvents in Organic Chemistry-1

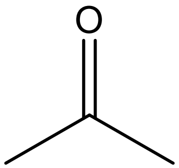
- Solvents should be “unreactive” (affect rates)
- Protic (H-bonding) or Aprotic
- Polar ($\varepsilon > 20$) or Nonpolar ($\varepsilon < 20$)

$$E = k\frac{q_1 q_2}{\varepsilon r}$$

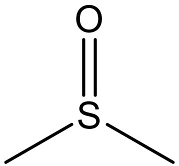
- S_N1 favors polar protic solvents (H_2O, ROH)
- S_N2 favors polar aprotic solvents



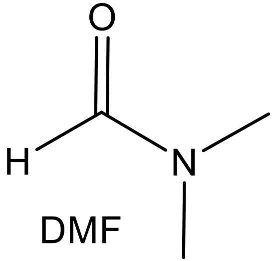
acetonitrile



acetone

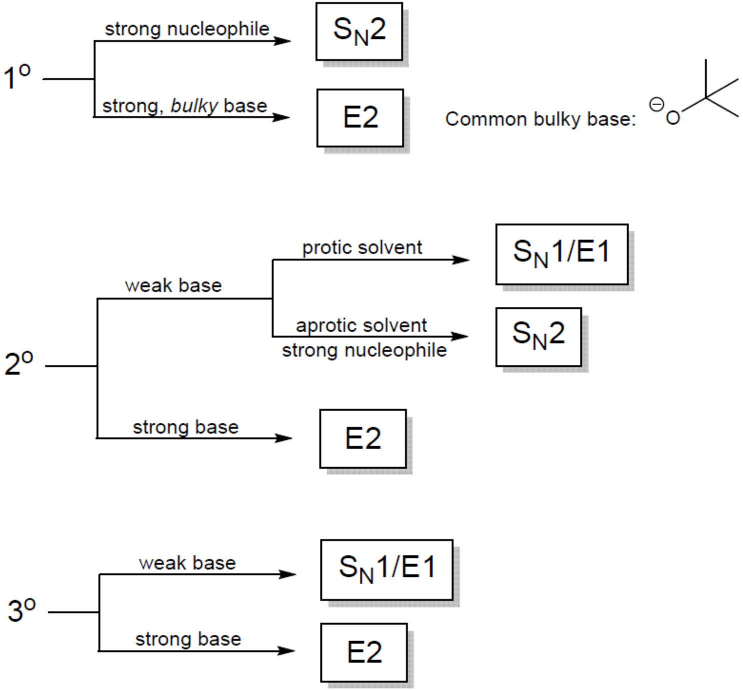


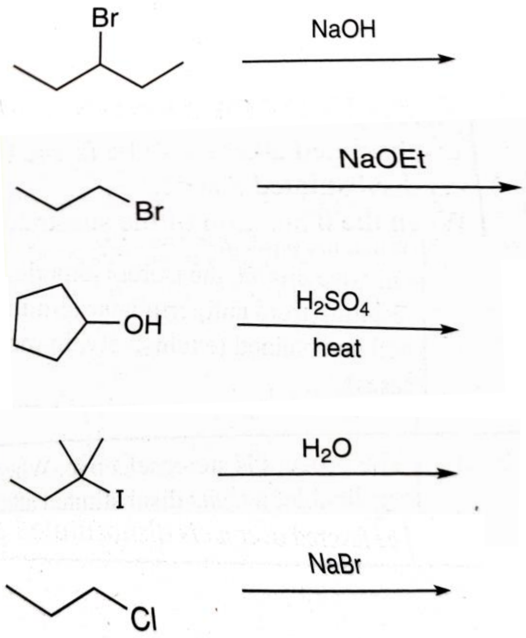
DMSO



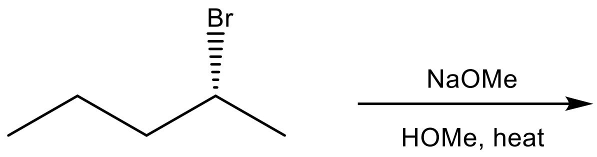
DMF

Identify the mechanism by which the major product will be formed for each of the following reaction.

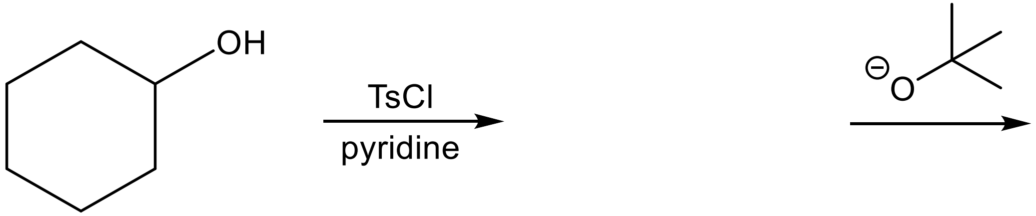




Predict products- identify the major and minor products

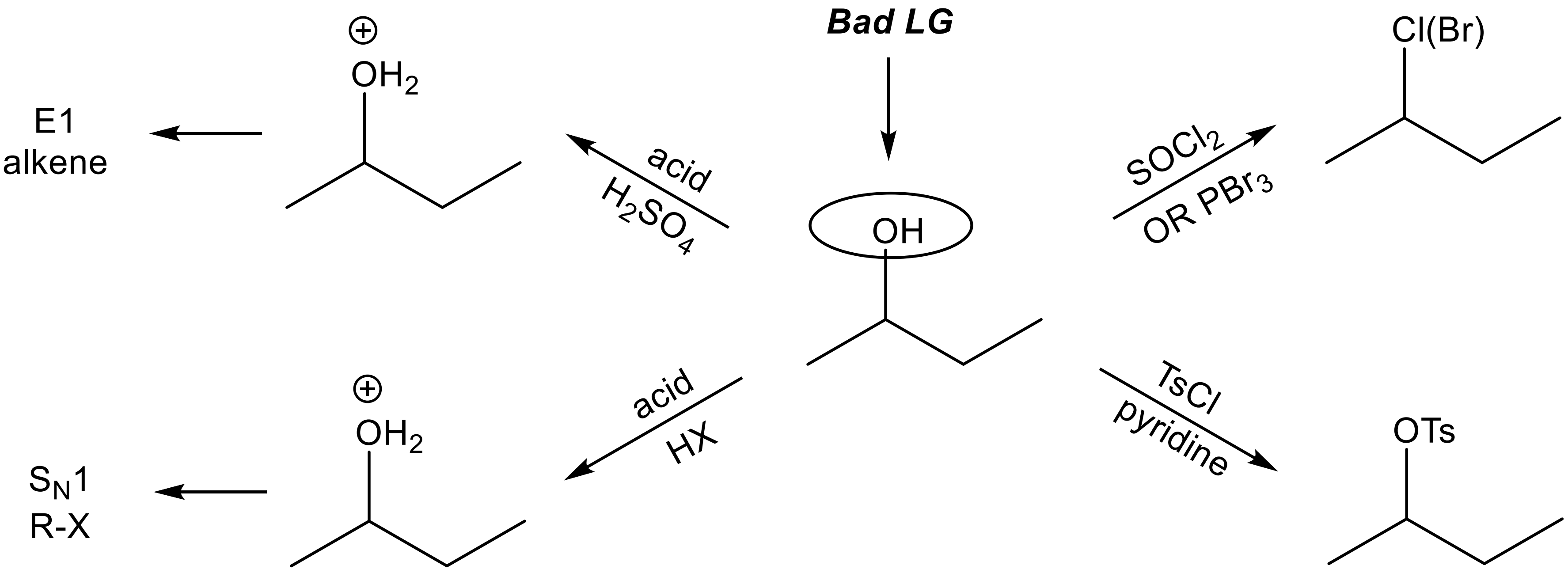


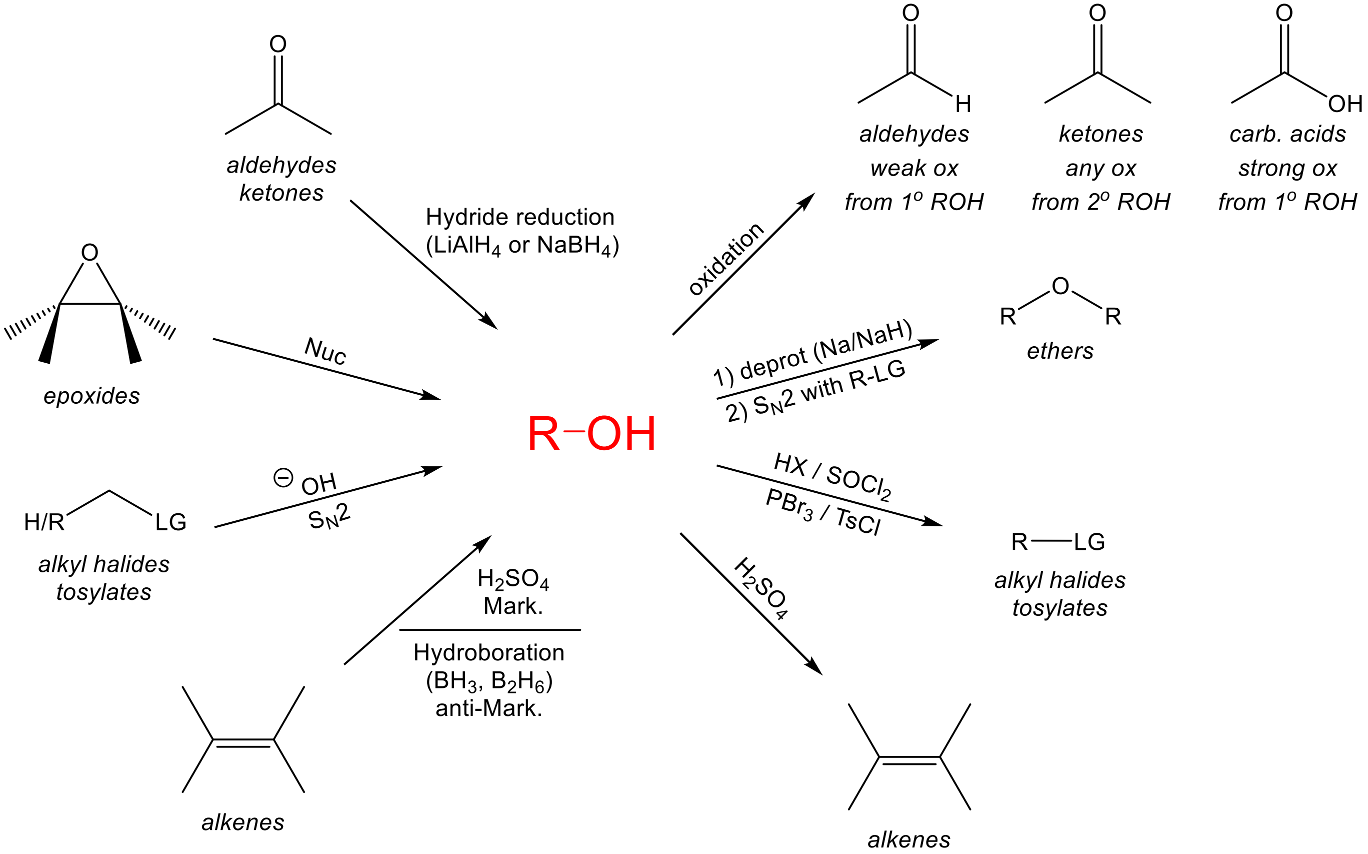
Predict products- identify the major and minor products

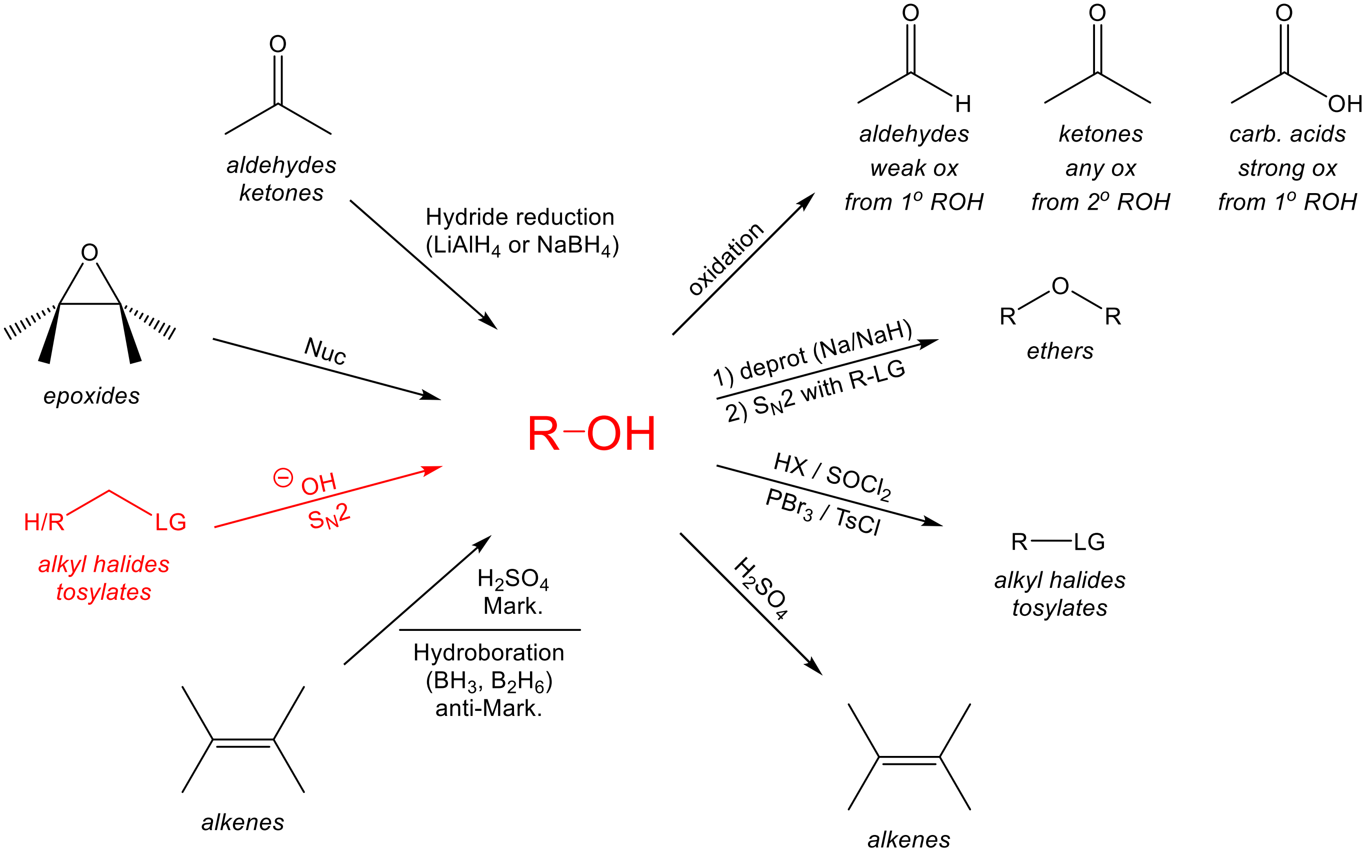


Predict products- identify the major and minor products

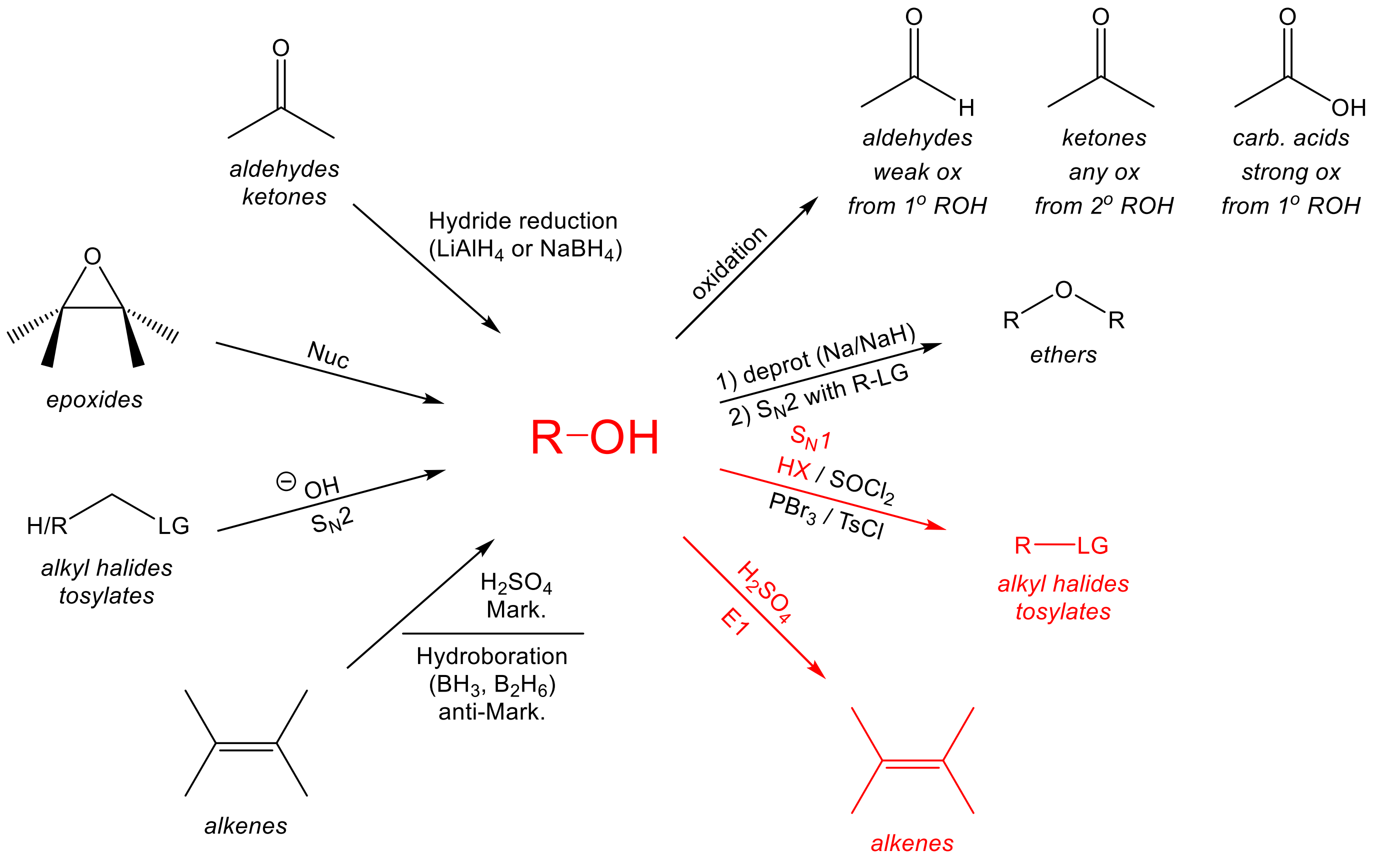
Alcohol Reactivity Overview




aldehydes
ketones
Hydride reduction
($LiAlH_4$ or $NaBH_4$)
epoxides
Nuc
alkyl halides
tosylates
H/R
LG
OH
S_N2
alkenes
H_2SO_4
Mark.
Hydroboration
(BH_3, B_2H_6)
anti-Mark.
R–OH
oxidation
aldehydes
weak ox
from 1° ROH
ketones
any ox
from 2° ROH
carb. acids
strong ox
from 1° ROH
1) deprot (Na/NaH)
2) S_N2 with R-LG
ethers
HX / $SOCl_2$
PBr_3 / TsCl
R—LG
alkyl halides
tosylates
H_2SO_4
alkenes



O
aldehydes
ketones
Hydride reduction
(LiAlH4 or NaBH4)
O
epoxides
Nuc
H/R
LG
alkyl halides
tosylates
OH
SN2
H2SO4
Mark.
Hydroboration
(BH3, B2H6)
anti-Mark.
alkenes
R–OH
oxidation
O
H
aldehydes
weak ox
from 1o ROH
O
ketones
any ox
from 2o ROH
O
OH
carb. acids
strong ox
from 1o ROH
1) deprot (Na/NaH)
2) SN2 with R-LG
R
O
R
ethers
HX / SOCl2
PBr3 / TsCl
R
LG
alkyl halides
tosylates
H2SO4
alkenes



aldehydes
ketones
Hydride reduction
(LiAlH4 or NaBH4)
epoxides
Nuc
H/R
LG
alkyl halides
tosylates
OH
SN2
R–OH
alkenes
H2SO4
Mark.
Hydroboration
(BH3, B2H6)
anti-Mark.
oxidation
H
aldehydes
weak ox
from 1o ROH
ketones
any ox
from 2o ROH
OH
carb. acids
strong ox
from 1o ROH
R
O
R
ethers
1) deprot (Na/NaH)
2) SN2 with R-LG
SN1
HX / SOCl2
PBr3 / TsCl
R
LG
alkyl halides
tosylates
H2SO4
E1
alkenes

Fill in the following with the major products